CASE STUDY: HIGH-CONCURRENCY E-VOTING ARCHITECTURE (PEMILWA UNY)

Role:	Lead Full-Stack Developer
Status:	Live / Production (2020 – Present)
Live URL:	https://pemilwa.uny.ac.id
Video Walkthrough:	https://youtu.be/boybejLOmRI

1. Project Overview

Pemilwa UNY is the official university-wide digital election platform. It serves thousands of active students, allowing them to authenticate and cast their votes securely. The system was engineered specifically to handle massive traffic spikes during peak election hours without downtime.

2. Technology Stack & Infrastructure

To maximize speed and handle high concurrency without the overhead of heavy backend frameworks, the system was built with a lightweight and highly optimized stack:

- **Backend & Database:** Native PHP with strict MySQL/PostgreSQL transaction protocols.
- **Authentication:** UNY Single Sign-On (SSO) integration via CAS (Central Authentication Service).
- **Web Server & Reverse Proxy:** Nginx 1.18.0 (configured as a reverse proxy to efficiently route traffic and manage concurrent connections).
- **Frontend UI:** Bootstrap 5.3.2 coupled with Bootstrap Icons 1.11.2 for a responsive, mobile-first voter experience.
- **Asset Optimization:** jsDelivr CDN (offloading static assets to reduce server load).
- **Media Integration:** Native YouTube Player embeds for real-time voter tutorials and guidelines.

3. Key Engineering Challenges & Solutions

A. Identity Verification & Security (SSO Integration)

The Challenge: Verifying the identity of thousands of active students securely without duplicating the university's sensitive credential database or risking data leaks.

The Solution: I integrated the platform directly with UNY's official Single Sign-On (SSO) using the CAS protocol. This ensures that the e-voting system never stores or processes user passwords. Authentication is

handled entirely by the university's central identity provider, significantly reducing the security attack surface and providing a seamless login experience for students.

B. Race Conditions & High-Concurrency (Traffic Spikes)

The Challenge: During the opening hours of the election, thousands of students log in and submit their ballots simultaneously. A poorly optimized system could result in race conditions, leading to double-voting or corrupted ballot data.

The Solution: By utilizing Native PHP, I bypassed the memory overhead of traditional frameworks. I implemented strict database locking mechanisms (ACID-compliant transactions). Furthermore, Nginx 1.18.0 was utilized as a reverse proxy to manage the connection queues, ensuring that even under extreme load, every single vote is processed sequentially and securely. The system has run flawlessly with 0% data corruption since 2020.

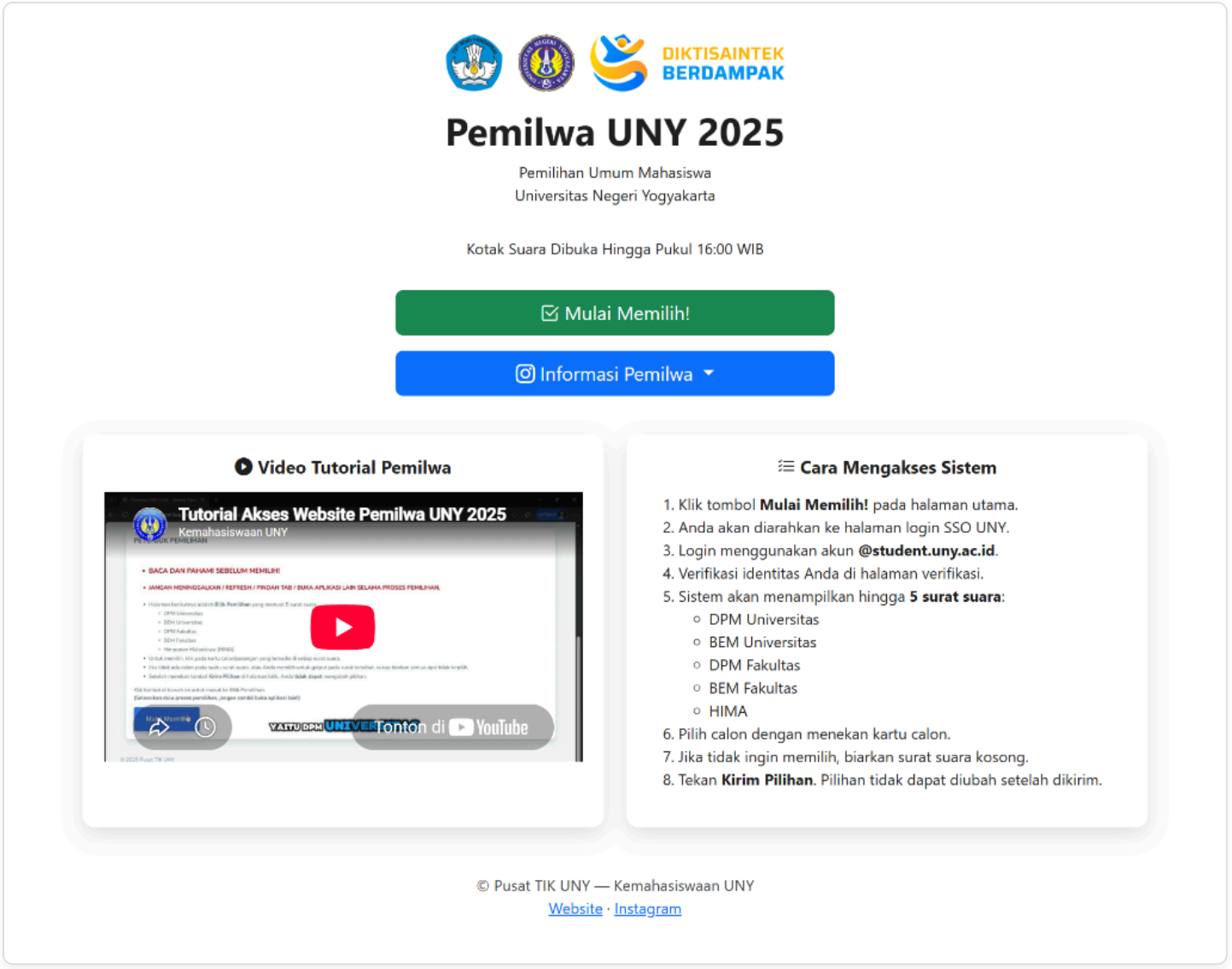
C. Data Privacy & Voter Anonymity

The Challenge: Ensuring that a student can only vote once, while guaranteeing that their actual ballot choice cannot be traced back to their student ID (NIM).

The Solution: Separation of concerns at the database level. The authentication table (which tracks who has voted via SSO) is completely isolated from the ballot table (which tracks what was voted for).

4. System Interface & User Flow

(Please refer to the embedded YouTube tutorial link above for a complete walkthrough of the internal system).



Homepage and Native YouTube Video Tutorial Embed (pemilwa.uny.ac.id)



SSO CAS Authentication Gateway for highly secure Identity Verification

Pemilwa UNY

BEM UNIVERSITAS

Kode Fakultas: **0** | Nomor Surat: **2**

DPT
39.725

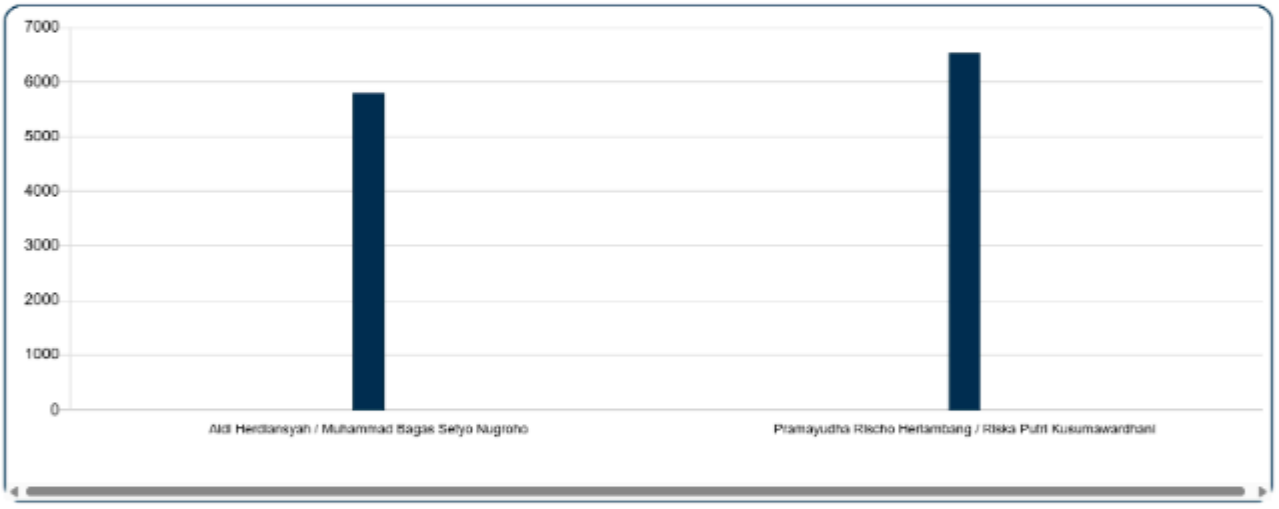
MEMILIH
12.331

TIDAK MEMILIH
27.394

TOTAL SUARA KANDIDAT
12.331

Diagram (Bar)

Pada layar mobile, diagram batang dapat terlihat kurang optimal. Disarankan membuka halaman ini menggunakan mode desktop.



Tabel

TINGKAT	NOMOR URUT	NAMA	SUARA
BEM UNIVERSITAS	1	**Ketua:** Aldi Herdiansyah **Wakil:** Muhammad Bagas Setyo Nugroho 	5805
BEM UNIVERSITAS	2	**Ketua:** Pramayudha Rischo Herlambang **Wakil:** Riska Putri Kusumawardhani 	6526

Kembali

Inside the E-Voting Dashboard – Live Voting Interface & Candidates

Live Count Jumlah Pemilih per Fakultas
Pemilwa UNY 2025

Halaman ini menampilkan **jumlah pemilih yang sudah terekam saat ini**. Data diperbarui otomatis setiap **5 detik**.

Waktu pembaruan: 09 Sep 2026, 13:55:24 WIB
Jika tampilan tabel terpotong di ponsel, geser ke samping (horizontal).

TOTAL PEMILIH SAAT INI	TOTAL DPT	PERSENTASE PARTISIPASI
13.807	**39.725**	**34,76%**

NO	FAKULTAS	JUMLAH PEMILIH	TOTAL DPT	PERSENTASE
1	**Fakultas Ilmu Sosial dan Ilmu Politik**	2.345	5.119	45,81%
2	**Fakultas Ekonomi dan Bisnis**	2.088	4.126	50,61%
3	**Fakultas Teknik**	1.674	6.068	27,59%
4	**Fakultas Matematika dan IPA**	1.605	3.789	42,36%
5	**Fakultas Ilmu Pendidikan**	1.585	4.770	33,23%
6	**Fakultas Bahasa, Seni, dan Budaya**	1.565	6.524	23,99%
7	**Fakultas Vokasi**	1.457	4.773	30,53%
8	**Fakultas Ilmu Keolahragaan dan Kesehatan**	800	3.170	25,24%
9	**Fakultas Psikologi**	353	992	35,58%
10	**Fakultas Hukum**	215	272	79,04%
11	**Fakultas Kedokteran**	120	122	98,36%
Total		**13.807**	**39.725**	**34,76%**

Catatan: angka partisipasi dihitung dari **pemilih terekam saat ini** dibandingkan total DPT.

Real-Time Quick Count & Participation Monitoring Dashboard